

Explainable AI-Based Digital Twin for PAUT Weld Inspection

1. Project Overview: This project focuses on the development of an intelligent Digital Twin framework for weld inspection using Phased Array Ultrasonic Testing (PAUT) integrated with Artificial Intelligence (AI) and Explainable AI (XAI) techniques.

The internship is designed as a multidisciplinary team-based project involving:

- NDT/PAUT domain knowledge
- Signal and image processing
- AI/ML model development
- Explainable AI techniques
- Digital Twin visualization
- Dashboard and UI integration

The proposed system aims to automatically detect weld defects from PAUT data while providing interpretable AI predictions and real-time visualization of weld health through a Digital Twin environment.

2. Recommended Team Size: 3-4 students

Duration 8–10 Weeks Summer Internship Program

3. Suggested Role Distribution

Student	Responsibility
Student 1	PAUT data acquisition, preprocessing, signal analysis
Student 2	AI/ML model development for defect detection
Student 3	Explainable AI (Grad-CAM, SHAP, interpretability)
Student 4	Digital Twin dashboard, visualization, integration

4. Milestones

1	NDT & Data Engineer • PAUT data collection • signal preprocessing • Defect annotation and dataset preparation
2	AI/XAI Engineer • CNN/YOLO model • explainability maps • performance analysis
3	Digital Twin Engineer • weld health visualization • dashboard • integration layer

5. Background

Phased Array Ultrasonic Testing (PAUT) is widely used in industrial Non-Destructive Testing (NDT) for inspection of welds and critical structures in sectors such as oil & gas, aerospace, power plants, and manufacturing. Conventional PAUT inspection requires experienced operators for defect interpretation, which can lead to subjectivity and inconsistency. Recent advancements in Artificial Intelligence (AI) and Digital Twin technologies provide opportunities to develop intelligent, real-time, and interpretable inspection systems. This project aims to develop an AI-driven Digital Twin framework for weld inspection using PAUT data integrated with Explainable AI (XAI) techniques. The system will not only detect defects automatically but also provide reasoning behind AI predictions, improving trust and reliability in safety-critical applications.

6. Objectives

The primary objectives of the project are:

1. To develop an AI/ML model for automated defect detection and classification using PAUT data.
2. To create a Digital Twin representation of weld health and defect status.
3. To integrate Explainable AI (XAI) methods for interpretation of AI-based decisions.
4. To visualize inspection results and defect severity through an interactive dashboard.

7. Proposed Methodology

The project will begin with collection and preprocessing of PAUT inspection data such as A-scan, B-scan, and TFM signals/images. Signal processing and image enhancement techniques will be applied to improve data quality.

Machine Learning and Deep Learning models such as CNN, Random Forest, or YOLO will be trained to identify weld defects including cracks, porosity, slag inclusion, and lack of fusion.

To improve interpretability, Explainable AI (xAI) techniques such as Grad-CAM and SHAP will be implemented to highlight critical regions and features responsible for defect prediction.

A Digital Twin model of the weld/component will then be developed to represent:

- weld geometry,
- defect locations,
- inspection history,
- health status, and
- defect severity.

The final system will include a visualization dashboard for monitoring weld condition and AI explanations.

8. Expected Deliverables

- AI-based defect detection model for PAUT inspection
- Explainable AI visualization module

- Digital Twin prototype for weld monitoring
- Interactive inspection dashboard
- Technical report and presentation

9. Tools & Technologies

- Python and its libraries; Streamlit/Dash; SHAP / Grad-CAM libraries

10. Expected Outcomes

The project will provide hands-on exposure to advanced NDT technologies, Artificial Intelligence, Digital Twin systems, and industrial inspection workflows. The developed framework can serve as a foundation for intelligent structural health monitoring systems and future research in trustworthy AI for NDT applications.

Possible Additional modules:

- defect growth prediction
- remaining useful life (RUL)
- FEM simulation
- real-time streaming
- cloud dashboard
- IoT integration

Final Technical Outcomes

At the end of the internship, the student team is expected to deliver the following consolidated outputs:

1. Source Code Repository

A well-structured and documented Git-based repository containing:

- PAUT data preprocessing pipeline
- AI/ML defect detection models
- Explainable AI (XAI) modules
- Digital Twin visualization framework
- Dashboard/UI implementation
- Model training and evaluation scripts
- Deployment and execution instructions

The repository should follow proper software engineering practices including:

- modular code structure,
- version control,
- documentation/comments,
- requirements file/environment setup, and
- reproducibility guidelines.

2. Technical Project Report

A comprehensive technical report documenting:

- problem statement and objectives,
- literature review,
- PAUT inspection methodology,
- dataset preparation and preprocessing,
- AI/ML model development,
- Explainable AI implementation,
- Digital Twin architecture,
- experimental results and performance metrics,
- challenges and limitations, and
- future scope and recommendations.

The report should include:

- workflow diagrams,
- model architecture figures,
- visualization outputs,
- result comparisons, and
- references to relevant research work.

3. Final Demonstration & Presentation

A final demonstration showcasing:

- automated defect detection,
- explainability visualization,

- Digital Twin monitoring dashboard, and
- integrated system workflow.

Students will also present:

- methodology,
- implementation details,
- results,
- technical learnings, and
- future enhancement opportunities.